Data Analytics Project: Railway Gauge Trends

Dataset Used: railway_gauge.csv

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This document outlines a structured, end-to-end data analysis project based on railway gauge data.

Dataset: Railway Gauge Data (Assumed to be in a CSV file)
Project Goal: To analyze historical trends and composition of different railway gauges (Broad, Metre, Narrow) over the years.**Modules Used**

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (np.diff).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling, Type Conversion
Pandas Operations	Filtering, Selection, Aggregation, Feature Engineering
NumPy Calculations	Yearly growth calculation (np.diff)
Visualization	Line Graph, Bar Chart, Pie Chart, Stacked Bar Chart

Scenario 1: Basic Data Loading & Cleaning (Easy)

Description:
In this Scenario,
The dataset is loaded into a Pandas DataFrame and inspected using the first few rows and column names. Missing values are handled by replacing them with zero, and all gauge columns are converted to numeric types for accurate analysis.

Task	Description
1.	Load the dataset into a Pandas DataFrame.
2.	Display the first 5 rows and column names to understand structure.
3.	Check for missing values and replace them with 0 (assuming missing values represent 0 km of track).
4.	Convert all relevant gauge columns (Broad, Metre, Narrow, Total) to numeric types (e.g., float64 or int64).

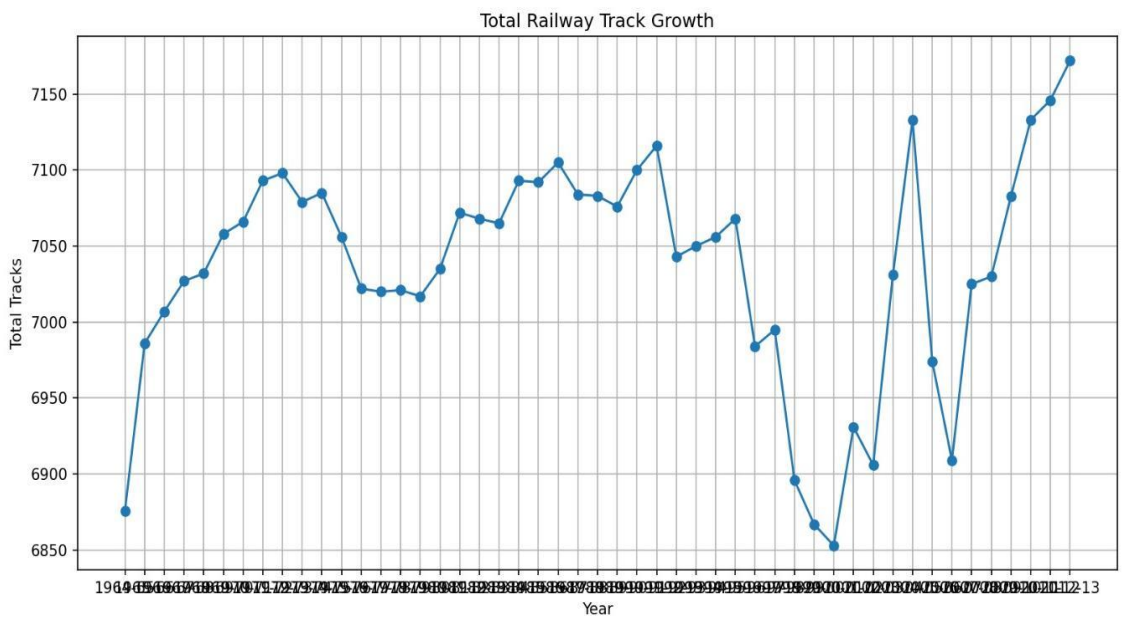
Conclusion:
The dataset was successfully loaded and cleaned for analysis. Missing values were handled, and all gauge columns were converted to numeric format. This ensures the data is accurate and ready for further processing.

Scenario 2: Simple Visualization (Easy)

Description:
In this scenario,
The Year and Total columns are used to plot a line graph showing railway growth over time. This helps in identifying whether the overall trend is increasing or decreasing.

Task	Description
1.	Extract the Year and Total columns.
2.	Plot a line graph showing **Total** tracks over the **Years**.
3.	Add a descriptive title (e.g., "Growth of Total Railway Network Over Time") and X/Y axis labels.
4.	Identify whether the overall trend is increasing or decreasing and note the finding.

Graph:



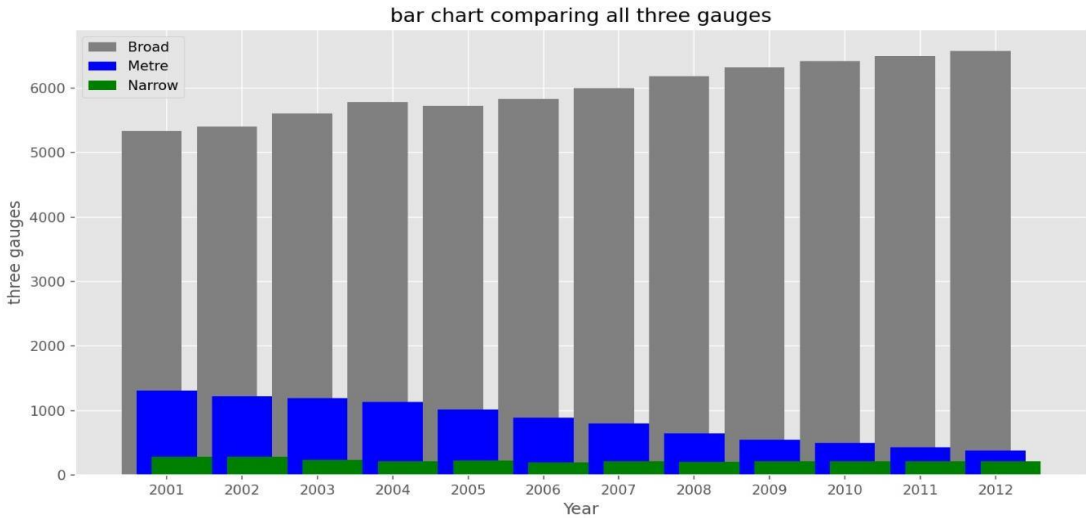
Conclusion:
The line graph shows an overall increasing trend in total railway tracks over the years. This indicates steady growth and expansion of the railway network.

Scenario 3: Filtering + Bar Chart (Medium)

Description:
In the scenario,
The dataset is filtered for years after 2000 and gauge types are compared using a grouped bar chart. This helps in identifying the dominant gauge in recent years.

Task	Description
1.	Filter the dataset to include only years *after* 2000.
2.	Select the Broad Gauge, Metre Gauge, and Narrow Gauge columns for the filtered data.
3.	Plot a **grouped bar chart** comparing the three gauges for the selected years.
4.	Add a legend and ensure proper labels for axes.
5.	Identify which gauge dominates in recent years (post-2000).

Graph:



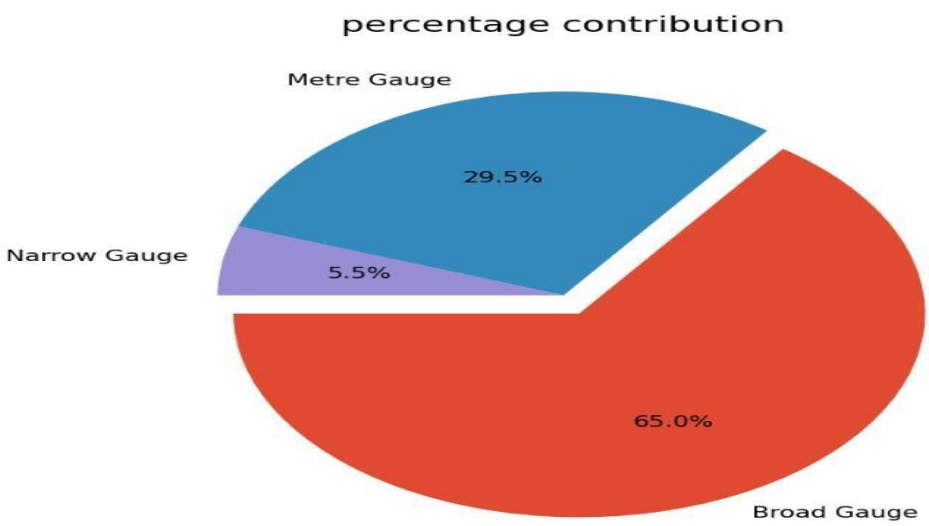
Conclusion:
The bar chart reveals that Broad Gauge dominates in recent years. Metre and Narrow Gauges have declined, indicating a shift towards standardization.

Scenario 4: Feature Engineering + Pie Chart (Medium)

Description:
In this scenario,
The total contribution of each gauge is calculated and visualized using a pie chart. This shows the percentage share of each gauge in the railway network.

Task	Description
1.	Calculate the total sum of each gauge (Broad Gauge, Metre Gauge, Narrow Gauge) across all available years.
2.	Create a new structure (Pandas Series or single-column DataFrame) holding these three total values.
3.	Plot a **pie chart** using this new structure to show the percentage contribution of each gauge type.
4.	Add percentage labels to the slices using autopct (e.g., '%.1f%%').
5.	Interpret which gauge contributes the most to the entire history of the railway network.

Graph:



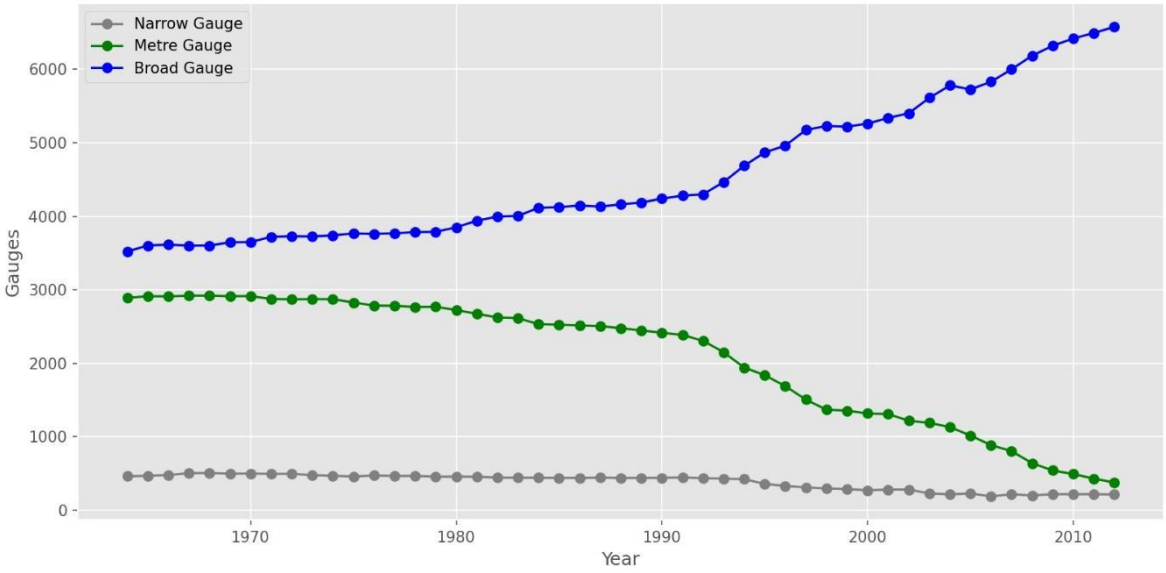
Conclusion:
The pie chart shows that Broad Gauge contributes the largest share to the railway network. Metre and Narrow Gauges have smaller contributions.

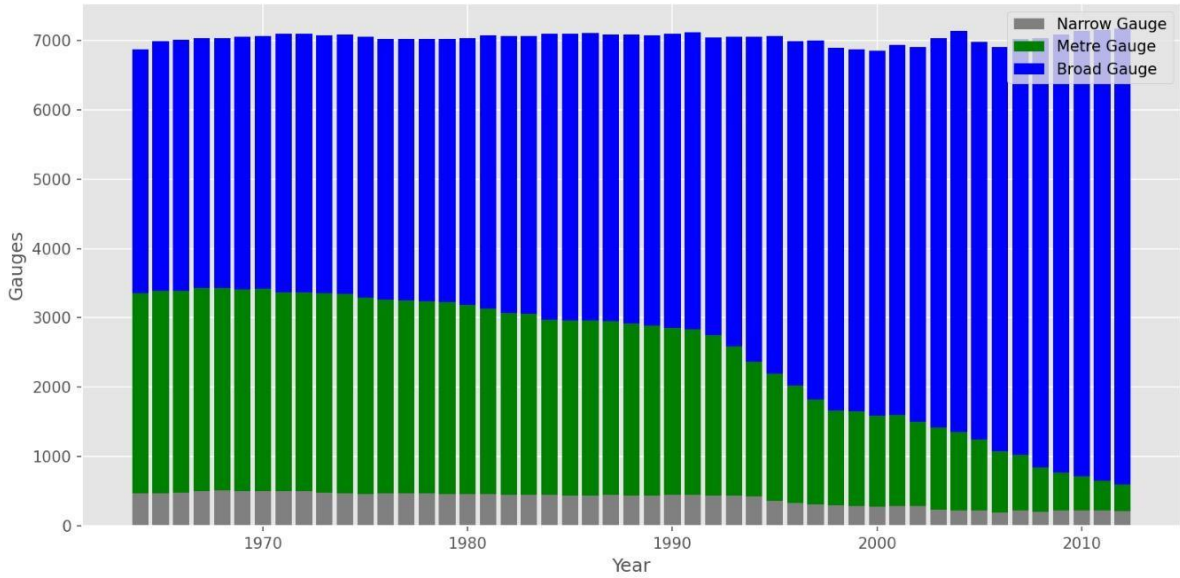
Scenario 5: Advanced Analysis + Multiple Graphs (Hard)

Description:
In this scenario,
The total contribution of each gauge is calculated and visualized using a pie chart. This shows the percentage share of each gauge in the railway network.

Task	Description
1.	**Create new columns** for percentage composition relative to Total tracks for each year: % Broad Gauge, % Metre Gauge, and % Narrow Gauge.
2.	Use the NumPy function np.diff to calculate the **yearly growth (change)** of the Total track column.
3.	Plot: a) A single **Line Graph** showing the trend of all three percentage composition columns (% Broad Gauge, % Metre Gauge, % Narrow Gauge) over the years. b) A **Stacked Bar Chart** showing the raw composition of the three gauges (Broad, Metre, Narrow) over time.
4.	**Highlight/Identify:** a) The year(s) with the highest yearly growth in Total tracks. b) Any specific decade or period showing a significant decline in any particular gauge.
5.	**Final Conclusion:** Based on the percentage composition line graph, provide a conclusion to the question: **"Is the railway system shifting towards a single dominant gauge?"**

Graph:





Conclusion:

The line graph shows that Broad Gauge is steadily increasing, while Metre and Narrow Gauges are declining over time. The **stacked bar chart** clearly highlights that Broad Gauge occupies most of the total track length in recent years. Together, both graphs indicate a strong shift towards Broad Gauge dominance. This confirms that the railway system is moving towards a single standardized gauge.

Final Conclusion:

The overall analysis shows that the railway network has consistently expanded over the years with a clear shift in gauge preference. Broad Gauge has increased significantly, while Metre and Narrow Gauges have declined. All visualizations confirm that Broad Gauge dominates in recent years. This indicates a move towards standardization for better efficiency and capacity. Hence, the railway system is evolving into a single dominant gauge system centered around Broad Gauge.

What you have learned?

In this project, I learned how to load and clean real-world data using Pandas, including handling missing values and converting data types. I also gained experience in filtering and analyzing data, and using NumPy for calculations like yearly growth. Additionally, I created different types of visualizations such as line graphs, bar charts, pie charts, and stacked charts. Finally, I learned how to interpret data trends and draw meaningful conclusions from analysis.